

Can LSTM Network outperform Tradition model in Forecasting Air pollutant Concentrations?

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Abstract— By applying a rolling-window framework, this study assesses the effectiveness of SARIMA, VAR, Holt-Winters, and LSTM models for short-term air pollution forecasting. RMSE was used to evaluate accuracy of forecasts over several time horizons, and standard deviation bands obtained from repeated window-based assessments were used to quantify uncertainty. The predictive models display cyclic RMSE fluctuations over the 72-hour horizon, indicating that pollutant-specific variations in time have a significant impact on prediction stability. While LSTM shows higher levels of variability but competitive performance during peak fluctuations, VAR and SARIMA achieve the lowest average RMSE values, indicating strong baseline consistency. Uncertainty bands emphasize the robustness and dependability of the framework, offering a more insightful assessment than single-value error metrics. All things considered, this multi-model analysis provides better understanding of comparative behavioral models and forecast stability.

Keywords- (SARIMA, VAR, Holt-Winter, LSTM, Rolling forecast, Urban pollution, Short-term prediction)

I. INTRODUCTION

Air pollutants are substances in the air that can harm human health or the environment when present in high concentrations. Air pollution has been a major issue that not only affects human health but also the environment. Human activities and meteorological factors strongly influence the presence of harmful pollutants such as PM_{2.5}, Nitrogen dioxide, sulfur dioxide, and ground-level ozone. Therefore, grasping how the air pollutants behave is crucial for making informed decisions that help to improve the lives of people (Marinov E. et al., 2022). Pollutants forecasting enables regulatory bodies to implement preventive measures and regulatory actions. However, accurate forecasting has been a challenging task due to the dynamic and volatile nature of air quality data.

This study uses time series analysis to forecast hourly concentrations of air pollutants. The study applies and compares various forecasting strategies (such as statistical models, exponential smoothing, multivariate, and machine learning) in an effort to find those that accurately forecast short and medium-term trends in air quality. The emphasis is on assessing four crucial strategies.

Research question

The research investigates the performance of traditional and machine learning in predicting hourly air pollutant concentrations over different forecast horizons. The proposed research aims to cover the following questions:

1. How does the forecasting accuracy of the long short-term memory (LSTM) model compare to that of traditional time series methods (SARIMA, Holt-Winters, and VAR) across different forecast horizons (1 to 72 hours)?
2. How does model performance degrade as the forecast horizon increases, and which approaches remain more robust over time?

Hypothesis

The following hypotheses and null hypotheses were developed to direct the model comparison and assess deep learning's efficacy in comparison to traditional time-series techniques:

- **(H₁):** LSTM will achieve lower forecasting error (RMSE) than traditional models (SARIMA, Holt-Winters, VAR), especially for longer forecast horizons (beyond 24 hours).
- **(H₀):** There is no significant difference or the Classical model outperforms in forecasting accuracy (RMSE) between LSTM and traditional time-series models across forecast horizons

Comparing and evaluating the traditional model and the machine learning best time series model based on performance for air pollutant forecasting is the main aim of this study. This enables us to comprehend the most suitable approaches for accurate, operational air quality forecasting in urban environments.

By presenting a methodical and equitable comparison between traditional time series models (SARIMA, Holt-Winters, and VAR) and contemporary deep learning models (LSTM) using actual hourly air pollutant data, this research contributes to the growing field of air quality forecasting. The report provides the strengths and limitations of multiple approaches under the same conditions and forecast horizons. Furthermore, the results will provide useful information on which models work well for both short-term and long-term projections, enabling the researchers and other environmental organizations to choose the best forecasting techniques.

Assessing the accuracy of multiple models by creating a forecasting pipeline using past hourly data on pollution

concentrations and weather variables is the proposed solution. Different models will be trained and adjusted to anticipate levels of pollutant for a maximum of 72 hours following extensive data preprocessing like handling missing values, removing outliers and negative values, and seasonal decompositions. Metrics, including RMSE, will be utilized to assess the forecast performance.

The need for data-driven environmental management plans to solve the growing problem of air pollution that depend on robust, interpretable, and operationally feasible forecasting techniques is what motivated this project. This report will provide a clear foundation of evidence for suitable forecasting models selection in urban air quality management by comparing different approaches of forecasting.

II. LITERATURE REVIEW

Forecasting air pollution has become a crucial part of environmental management, urban planning, and public health protection. Air pollution (like NO₂, SO₂, O₃, and PM_{2.5},) shows significant temporal and seasonal patterns (Agarwal, Shivang et al., 2020) that are impacted by weather conditions and human activity. Accurately forecasting short-term and medium-term pollution enables the authorities to provide early warning, a data-driven plan for traffic implementation or restrictions on industrial emissions, and to reduce the exposure of the population. The research paper (Agarwal, Shivang et al., 2020; Shi Yishe et al., 2021) emphasized the need for the model's capability to capture complex temporal dynamics while remaining robust and interpretable across different forecast horizons.

Traditional time series Forecasting techniques are still often employed because of their ease of use, interpretability, and well-established statistical characteristics. The univariate time series with distinct trends and seasonality components are easy to capture with models such as Seasonal autoregressive integrated moving average (SARIMA) and Holt-Winters exponential smoothing. The hourly and daily air pollutant data, especially with strong seasonal behavior, can be captured successfully using the SARIMA model (Necula Sabina-Cristiana et al., 2025). For short-term forecasting scenarios, Holt-Winters approaches, which assign exponentially decreasing weights to past data, have demonstrated reliable performance (Subhash/ & Kamlakar, 2025). Additionally, by capturing linear interdependencies across many time series, a multivariate technique like (Haslbeck Jonas M. B. et al., 2020) (VAR) enables the combined modeling of multiple pollutant and climatic variables (Haslbeck Jonas M. B. et al., 2020).

Air pollutant concentration usually shows irregular fluctuation caused by different factors such as weather patterns, human mobility, and other stochastic factors. The main drawback of traditional models such as SARIMA, HOLT_WINTER, and VAR is, these models cannot capture non-linearity and complexity of temporal dependencies. This has surged the demand for machine learning techniques, especially deep learning models like long short-term memory (LSTM) neural networks. The LSTM model is designed in a specific way so that it learns from sequential data and

exhibits promising results for enhancing forecasting accuracy for different pollutants such as PM_{2.5}, and O₃ (Malashin Ivan et al., 2024). According to a study (Brown Michael et al., 2024) contrasting LSTM with SARIMA and other classical models, LSTM typically performs better than conventional techniques for longer forecast horizons, when it becomes more important to capture temporal relationships.

The recent study has also promoted the significance of fair and systematic comparisons across various approaches. Although the classical statistical models are capable of handling the short-term and highly seasonal data, LSTM works better for multi-step forecasting because of its capability to manage non-linearity and complicated relationships (Agarwal, Shivang et al., 2020; Brown, Michael et al., 2024). The research question of the project, which evaluates the performance of different models such as SARIMA, JOLT-Winter, Var, and LSTM over a forecast horizon ranging from 1 to 72 hrs. and tests the hypothesis that LSTM will acquire less error than other conventional models for longer forecast horizons, is directly aligned with this.

Overall, the project in hand indicates that deep learning techniques are superior at capturing intricate temporal dynamics, even though statistical models are still useful for their interpretability and simplicity. By comparing model like SARIMA, HOLT-Winters, VAR, and LSTM based on their performance, structure, and multi-horizon on actual hourly pollutant data, the project at hand adds to the current body of research.

III. METHOD DESCRIPTION

A. The Dataset

The dataset used for the project at hand was obtained from Quito, Municipal Air Quality Monitoring Network, particularly from the Belisario monitoring station. The dataset consists of more than 15 years of data from 2004 to 2024. For the proposed project, we only used one complete recent year (Sept 2023- Aug 2024) to capture a full annual cycle while avoiding multi-year regime shifts (policy changes, instrumentation drift, pandemic effects). The forecasting targets are five key pollutants: NO₂ ug/m³, SO₂ ug/m³, CO mg/m³, O₃ ug/m³, and PM_{2.5}, ug/m³, which are standard indicators of urban air quality and demonstrate strong diurnal/seasonal structure. Additionally, the data also includes meteorological variables such as temperature, Rain, Solar radiation, relative humidity, wind speed, wind direction, and atm pressure, which influence pollutant dispersion and formation.

B. Preprocessing

To make sure the dataset was clean, continuous, and appropriate for time-series forecasting, a thorough and repeatable preparation pipeline was created prior to model training. To ensure fairness and comparability, every operation was predetermined and applied uniformly across all models.

First, the timestamp column (date_time) was converted to a datetime format and set as the index to ensure that data is not duplicated or has any missing timestamps. We convert all data to numeric types; also, any non-numeric or corrupted entries brought on by sensor failures were then replaced with missing values (NaN.) to ensure that the data is numeric and systematic handling of missing values.

Then, the missing value in the data set was handled. Meteorological variables were time-interpolated, as it preserves realistic transitions without introducing step changes. For pollutants, seasonal-trend decomposition (STL) with 24 hours was first applied to separate the data into seasonal, trend, and residual components. Then, missing values were handled using interpolation on the residual component before recombining it with the seasonal signal. Lastly, KNNImputer (k=5) was utilized to handle the remaining small gaps and preserve multivariate coherence across variables.

In order to eliminate sensor noise (outliers) while maintaining actual environmental variability, outlier treatment was then carried out. Physically unfeasible negative pollutant concentrations were trimmed to zero. Each pollutant series was Winsorized at the first and 99th percentiles to lessen the impact of extreme outliers brought on by sensor spikes. By capping extreme values rather than eliminating them, this technique preserves the overall framework of the dataset and keeps such points from skewing model training.

Lastly, before training the LSTM model, pollutant data were normalized using Min–Max scaling to a range of [0, 1]. In addition to ensuring that every variable contributed equally throughout training, this scaling stabilized model convergence. To maintain interpretability, the data were kept in their original physical units for conventional models as SARIMA, Holt-Winters, and VAR.

C. Modeling

Hourly air pollution concentrations were predicted up to 72 hours in advance using four distinct time-series forecasting models: SARIMA, Holt-Winters, Vector Autoregression (VAR), and Long Short-Term Memory (LSTM). The objective was to conduct a systematic comparison, under consistent experimental circumstances, between contemporary deep learning techniques and conventional statistical methodologies.

The hourly concentrations of pollutants were predicted using the SARIMA model as a traditional statistical baseline. It is suitable for data on urban air quality because it works effectively with time series that have strong daily or weekly cycles. The configuration SARIMA (2,0,2) (1,1,1,24) was used for modeling each pollutant separately. While the seasonal parameters (P, D, Q, 24) clearly depict the 24-hour daily cycle typical of pollution associated with traffic patterns, the non-seasonal parameters (p, d, q) show short-term autocorrelation. 72-hour predictions were produced after the model was fitted using maximum likelihood estimation for every rolling training window (six

months of hourly data). In order to prevent data leaking, the rolling window technique makes sure the framework adjusts to changes in time while upholding tight chronological order. Instead of adjusting numerous parameters, a predefined, well-established seasonal and non-seasonal sequence was used to reduce SARIMA overfitting. Additionally, the model is inherently regularized by the seasonal structure and differencing terms, avoiding needless complexity. SARIMA was chosen due to its interpretability, computational effectiveness, and reputation for performing well on seasonal, organized environmental time series. For comparison with more sophisticated models like LSTM, it offers a trustworthy baseline.

The trend and seasonal variations found in daily pollution concentrations were captured using the Holt-Winters model. It incorporates longer-term behaviour while emphasizing recent findings using exponentially weighted smoothing. The daily variation in pollution levels was reflected in this study using an additive trend and an additive 24-hour seasonal component. Exponential Smoothing from statsmodels was used to fit the model; it dynamically improved the smoothing parameters (α , β , and γ) by minimizing the prediction error. The algorithm produced 72-hour forecasts for each rolling window after being trained on hourly data from the preceding six months. Holt-Winters' exponential smoothing framework, in which the parameters α , β , and γ are tuned for equal responsiveness and smoothness, prevents overfitting. The model's additive structure makes it straightforward and guards against overfitting on transient noise. For pollutants like SO₂ and CO that show typical recurring daily cycles, Holt-Winters was selected because it is easy to understand, straightforward, and efficient. It offers a robust and computationally effective baseline, but it is not as good at handling abrupt spikes or nonlinear patterns as LSTM.

Associations among multiple pollutants were simultaneously captured using the VAR model. VAR treats each pollutant as possibly influenced by both its own prior values and the past values of other pollutants, in contrast to univariate models. The model in this study underwent training utilizing all five pollutant time series over the same six-month rolling windows as earlier approaches. Only stable models (with eigenvalues less than 1) were employed, and the Akaike Information Criterion (AIC) was utilized to automatically determine the ideal lag order for each window. The Akaike Information Criterion (AIC), which penalizes excessively complicated lag orders, was used to control overfitting risk for VAR. To further avoid unstable or overfitted solutions, only stable models (eigenvalues < 1) were kept. To provide 72-hour-ahead forecasts for every pollutant, forecasts were created iteratively. VAR was selected because it is capable of capturing co-movements and interdependencies, for example, the inverse relationship between NO₂ and O₃, offering a multivariate viewpoint that conventional univariate models are unable to. It maintains

computational efficiency while providing consistent and comprehensible projections.

The LSTM model can recognize long-range patterns and nonlinear temporal correlations in sequential data, so was chosen as the deep learning method. Traffic, weather, variations in emissions, and chemical interactions all have an impact on the complicated behaviours of air pollution levels. When interactions are nonlinear or when time dependencies go beyond basic seasonality, traditional models perform poorly. The gated memory structure of LSTM offers an effective framework for simulating such dynamics.

A univariate time series was utilized for every pollutant. Each rolling window (six months of hourly data) was standardized using Min–Max Scaling before training. To prevent data leaking, the scaler was only fitted to the training set. Normalization stabilized gradient flow and accelerated convergence by putting the data in the range [0,1].

The normalized series were transformed into supervised samples using a sliding-window technique. The model was able to learn daily cycles and multi-day temporal patterns by using a 48-hour look-back window as input for each training sample. As a result, training sequences of shape (samples, 48, 1) were generated. Training and validation sets were created using an 85/15 chronological split.

Two-layer LSTM layers with 64 units and 32 units each made up the model. The network can learn hierarchical temporal representations—broader temporal dynamics in the upper layer and short-term trends in the bottom layer—thanks to stacked LSTMs. In order to minimize internal variance shift and stabilize training, batch-level normalization layers were included.

Dropout layers (0.25 and 0.20) randomly deactivated certain neurons during training to reduce overfitting and improve the model's ability to generalize to new inputs. By lowering internal covariate shifts, batch normalization significantly stabilized learning and enhanced generalization. Before the final Dense (1) output neuron predicted the next-hour pollutant value, a Dense (32) layer with ReLU activation offered nonlinear transformation capability. We kept an eye on training and validation loss patterns throughout epochs in along with architectural safeguards to make sure the model stopped training before overfitting or divergence became apparent.

The Adam optimizer (learning rate 0.0005), which adaptively modifies learning rates for rapid convergence and improved handling of noisy variances, was used to train the model. Mean Squared Error (MSE), a common loss function for regression tasks since it penalizes greater errors more severely, was used to train the LSTM model. However, Root Mean Squared Error (RMSE) was employed throughout evaluation since it can be more interpretable for evaluating model performance over various pollutants and is given in the same physical units as the pollutant concentrations. Early Stopping monitored the validation loss

to avoid overfitting and halted training if no enhancement was found for five consecutive epochs. The model used an autoregressive forecasting technique to produce predictions after being trained for each rolling window. The model projected the concentration for the following hour based on the last 48 normalized observations from the training window. The prediction was then utilized as input to forecast the next hour and added to the series, substituting the oldest value. Until all 72 hourly forecasts were generated, this iterative procedure was carried out. In order to ensure that outputs were restored to their initial pollutant units for precise error calculation, anticipated outcomes were then inversely scaled using the same Min–Max scaler fitted on the training data.

Because it captures both nonlinear and long-range temporal dependencies, this method is effective. The model can adjust to seasonal changes and evolving pollution patterns thanks to the usage of rolling windows. While autoregressive forecasting offers a realistic simulation of the model's behavior in actual operational systems, dropout and early termination decrease overfitting.

Error bars, also known as shaded variability regions, were calculated based on the standard deviation of RMSE values gathered for every rolling windows in order to measure predicting uncertainty for each model. The RMSE was first computed for every window for every predicted lead time, and then the mean and standard deviation were obtained across windows:

- **Mean RMSE:** average forecasting error at each lead
- **Standard deviation (SD):** variation in the model's error values at that lead
- **Error bar range:** Mean RMSE $\pm$ SD

Plotting this spectrum shows how consistent or erratic each model's accuracy is as the forecast horizon lengthens. Wider bands show more uncertainty or sensitivity to changing conditions, whereas narrower bands show more reliable predictions.

IV. RESULTS AND ANALYSIS

In order to evaluate the model performance, Root Mean Squared Error (RMSE) was utilized, which evaluates how closely the predicted pollutant concentrations matched the observed values, as it penalizes larger deviations and is standard in air-quality forecasting. Forecasts were created 72 hours in advance for every rolling window, and RMSE was calculated independently for each pollutant and forecast horizon (1–72 hours). We were able to figure out both mean performance and variability thanks to the rolling-window structure's several independent forecast windows, which improved the results' statistical reliability.

Table 1: Average RMSE and standard deviation across all rolling-window forecasts for the four models (VAR, SARIMA, Holt-Winters, and LSTM), summarizing

overall predictive accuracy and variability in hourly air-quality forecasting performance.

Models	Average_RMSE	Std-deviation
VAR	6.641026	5.455265
SARIMA	6.675296	5.482234
Holt-Winters	7.554095	5.778898
LSTM	8.861107	7.087144

From Table 1, the average RMSE and standard deviation for all models. While the standard deviation measures the variation or stability of the model's results across multiple time segments, the mean RMSE reflects the overall forecast accuracy. A lower standard deviation indicates more consistent predicting behaviour. According to the findings, VAR and SARIMA have the lowest standard deviation and average RMSE, indicating stability and accuracy. While LSTM exhibits the highest standard deviation, indicating its sensitivity to shifting temporal conditions and insufficient training data, Holt-Winters performs moderately with somewhat higher variability. This demonstrates that while LSTM needs a greater historical depth, and different variables to accomplish stable learning of long-range relationships, classical models offer more reliable forecasts under a constrained one-year dataset.

The model comparison plots demonstrate how forecasting accuracy and variability alter with increasing lead time by evaluating each model under identical conditions. The figures show model accuracy, stability, and responsiveness to daily cycles or sudden pollutant changes by displaying RMSE and standard deviation over a 72-hour horizon for each pollutant. By emphasizing the model's advantages, disadvantages, and practical suitability for both short- and long-term forecasting, this visual evaluation enhances numerical results. The figures of Model comparison, given below, display leads(1-72) on the x-axis and mean RMSE on the y-axis.

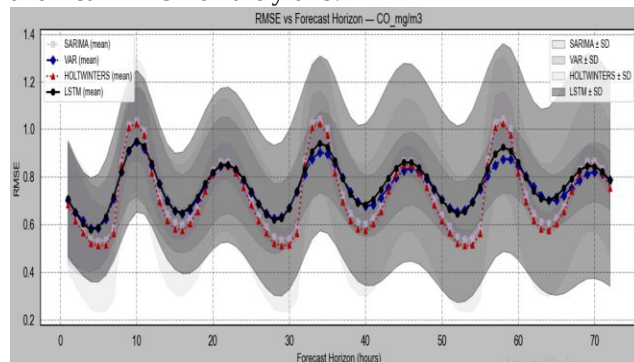


Figure 1 Forecast accuracy comparison for CO over a 72-hour horizon, showing mean RMSE (mg/m³) for SARIMA, VAR, Holt-Winters, and LSTM models versus forecast lead (hours), with ± 1 SD error bands reflecting uncertainty and the periodic RMSE fluctuations driven by the diurnal cycle in CO concentrations.

Compared to other pollutants, the CO prediction findings exhibit smoother RMSE changes, suggesting that the dynamics of CO levels are more predictable and less unpredictable. The uncertainty bands of all four models greatly overlap, and their mean RMSE courses are tightly linked. The tight confidence areas of SARIMA and VAR show that they preserve the least amount of variability. While LSTM once more displays the broadest shading, indicating greater inconsistency across rolling windows, Holt-Winters displays moderate uncertainty. Nevertheless, LSTM's mean RMSE trajectory stays near other models, demonstrating that LSTM can successfully track the pollutant trend despite greater unpredictability. With little variation in performance, CO is generally the least difficult pollutant for the models.

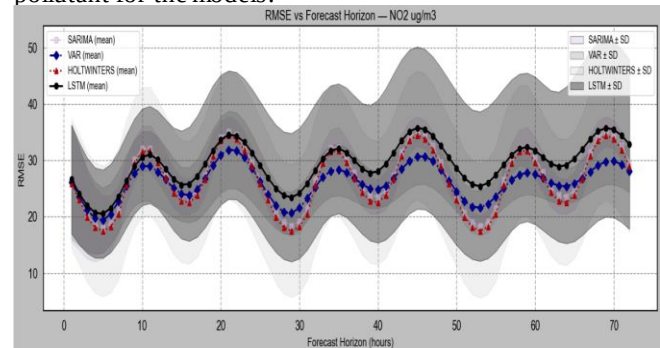


Figure 2 Forecast accuracy comparison for NO₂ over a 72-hour horizon, showing mean RMSE (µg/m³) for SARIMA, VAR, Holt-Winters, and LSTM models versus forecast lead (hours), with ± 1 SD error bands illustrating uncertainty and periodic fluctuations reflecting the diurnal cycle in NO₂ levels.

Figure 2 demonstrates the comparison of model performance for the NO₂ pollutant, forecast lead on the x-axis, and RMSE on y-axis. Strong diurnal behaviour is indicated by the NO₂ forecast evaluation, which displays a distinct cyclic pattern in RMSE across the 72-hour horizon. spike and dips are continually captured by all four models, which exhibit a similar pattern. Significant variations in model stability are evident in the shaded uncertainty bands: the largest is observed in LSTM, which may indicate overfitting to local fluctuations and increased sensitivity to window changes. Narrower bands are maintained by SARIMA and VAR, indicating more reliable and consistent forecasts. Holt-Winters, which falls between LSTM and classical models, exhibits moderate variability. Overall, NO₂ seems to be fairly predictable, but the predictive models' levels of uncertainty clearly vary, with VAR and SARIMA showing the most reliable results.

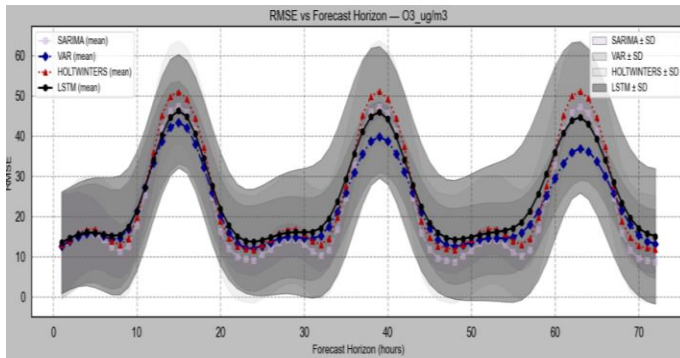


Figure 3 Forecast accuracy comparison for O_3 over a 72-hour horizon, showing mean RMSE ($\mu\text{g}/\text{m}^3$) for SARIMA, VAR, Holt-Winters, and LSTM models versus forecast lead (hours), with ± 1 SD error bands illustrating forecast uncertainty and periodic RMSE fluctuations driven by the diurnal ozone cycle.

Figure 3 exhibits Strong periodic spikes that correlate to recurring diurnal ozone cycles are visible in the O_3 forecasting plot. These cyclical peaks, which take place around comparable forecast horizons, are captured by all models. Near peaks, where the levels of ozone tend to move more erratically, the shaded error zones grow noticeably broader, suggesting a higher degree of intrinsic uncertainty in the pollutant behavior. Once more, LSTM exhibits the greatest variability, particularly during climax cycles, indicating the challenge of reliably forecasting high-variability segments. Even during peak times, SARIMA and VAR maintain smaller uncertainty bounds, indicating greater robustness. In the middle, Holt-Winters captures periodic structure with a modest degree of variability. Ozone's strong nonlinear variations make it the most difficult pollutant to anticipate overall.

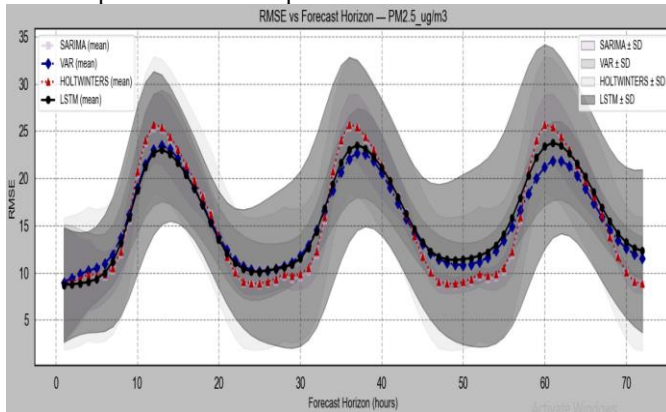


Figure 4 Forecast accuracy comparison for $PM_{2.5}$ over a 72-hour horizon, showing mean RMSE ($\mu\text{g}/\text{m}^3$) for SARIMA, VAR, Holt-Winters, and LSTM models versus forecast lead (hours), with ± 1 SD error bands capturing uncertainty and reflecting the strong diurnal cycles that drive periodic RMSE fluctuations.

Forecasts for $PM_{2.5}$ exhibit RMSE oscillations that are clearly periodic yet rather smooth. The main peaks and dips

are successfully captured by all models, although LSTM's uncertainty band is substantially thicker, indicating variation in performance within various windows. whereas, narrower dark regions are consistently produced by SARIMA and VAR, demonstrating consistent results even when RMSE values rise. Compared to VAR, Holt-Winters exhibits a somewhat greater degree of fluctuation. Although the wider LSTM uncertainty indicates a vulnerability to variable particulate matter episodes that classical models manage more reliably, the continuous periodicity in $PM_{2.5}$ behaviour enables every model function quite well.

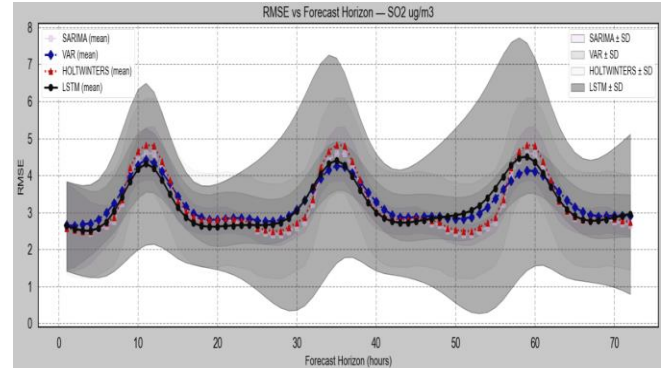


Figure Forecast accuracy comparison for SO_2 over a 72-hour horizon, showing mean RMSE ($\mu\text{g}/\text{m}^3$) for SARIMA, VAR, Holt-Winters, and LSTM models versus forecast lead (hours), with ± 1 SD error bands illustrating uncertainty and periodic fluctuations reflecting the diurnal cycle in SO_2 levels.

When compared to other pollutants, SO_2 forecasts exhibit smoother fluctuations and smaller RMSE ranges, indicating that SO_2 time series are less erratic and simpler to forecast, is demonstrated in Figure 5. There is really little difference between LSTM and traditional approaches in the mean RMSE curves of all models. Particularly for VAR and SARIMA, which exhibit tight, steady performance over the horizon, the uncertainty bands among models continue to be lower than in other pollutants. Once more, LSTM displays the largest SD band, emphasizing increased irregularity. Holt-Winters has modest variability but follows a similar trajectory. This suggests that while classical models continue to be more stable, a more straightforward pollutant profile enables all models to function equally.

The comparatively narrow shaded error bands representing standard deviation indicate that classical time-series models (SARIMA, VAR, Holt-Winters) work more consistently for every pollutant. Wider uncertainty envelopes are often the result of LSTM's higher RMSE variability. The model's sensitivity to variations in training windows and propensity to overfit local fluctuations are reflected in this behaviour. Despite this, LSTM frequently does a good job of tracking the mean trend, but it is inconsistent across forecasts. Instead of concentrating only on average RMSE curves, error bands offer crucial information about model reliability, highlighting the significance of incorporating uncertainty visualization.

Discussion

The RMSE values displayed for each prediction horizon are averages calculated over several rolling windows to increase the statistical accuracy of the findings. Similar to cross-validation in time-series settings, each window functions as a repeated experiment. Model uncertainty is reflected in the variation in RMSE across windows, particularly for longer prediction horizons where fluctuations are more noticeable. The rolling evaluation naturally captures performance variability across various temporal contexts (weekdays, weekends, seasonal shifts), even when explicit error bars are not displayed.

The LSTM model showed signs of overfitting, especially after 24 hours, when its RMSE increased more quickly than that of classical models. This implies that even with dropout and early termination, the LSTM was partially overfitting the small amount of training data. Due to their more straightforward structure and greater dependence on seasonal patterns, classical models continued to be more stable.

From a limit's standpoint, complicated models like LSTM may have had less predictive power due to the limited one-year dataset, the lack of other meteorological factors and timestamp variables like hour of the day, day of the week. Abrupt spikes in pollution were a problem for all models, as seen by recurrent peaks in RMSE around 24-hour intervals. These are abrupt, hard-to-predict fluctuations in emissions.

Future developments might incorporate hybrid deep-learning models, train on multi-year data, add meteorological variables, and quantify uncertainty via bootstrapping or error bars. Under actual operating conditions, these improvements would probably increase robustness and enable stronger generalization.

V. CONCLUSION

In the current research, four forecasting techniques for short-term (1–72 hour) air pollution prediction have been compared: SARIMA, Holt-Winters, VAR, and LSTM. According to the analysis, the most consistent results over forecast horizons and the lowest average RMSE were attained by traditional statistical models, especially VAR and SARIMA. Since the research utilized only one year of training data and inadequate modelling of long-range dependencies, LSTM was able to capture complex temporal dynamics but showed higher uncertainty and increasing error at longer horizons. For pollutants with strong daily cycles, Holt-Winters reliably performed well.

The deep-learning method successfully caught complicated temporal dynamics and showed promise for greater accuracy with larger datasets and longer historical windows, even though it did not outperform traditional models in this context. The result show that pollutant dynamics, data volume, and data quality all have a significant impact on model appropriateness. For short-term

forecasting, classical models continue to be computationally effective and competitive.

By showing that deep-learning techniques need richer data and longer historical windows to consistently outperform classical models for short-term pollutant forecasting, these results directly address the research question. Model performance was impacted by constraints like pollutant-only inputs, limited temporal context, and partial overfitting. In addition to probabilistic techniques for uncertainty quantification, future research should include weather parameters, larger historical datasets, and sophisticated architectures like hybrid or focus-based models.

In general, the findings show how data features, model complexity, and forecast stability interact, providing significant data for choosing suitable forecasting methods in daily air-quality management.

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